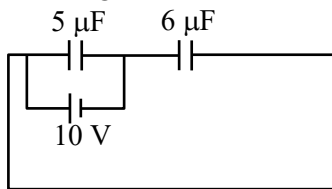
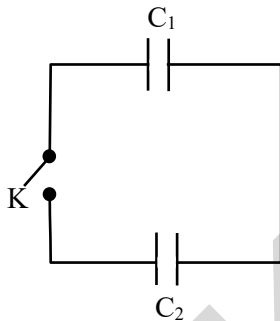


PHYSICS

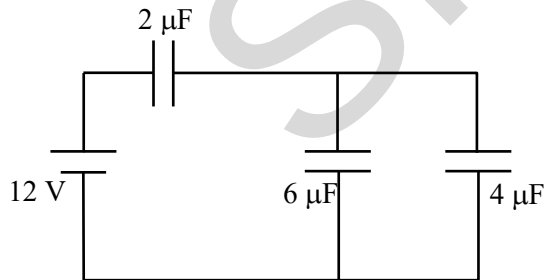
1. The charge supplied by the battery in the circuit shown in the figure



- (1) $\frac{300}{11} \mu C$ (2) $60 \mu C$
 (3) $50 \mu C$ (4) $110 \mu C$
2. How much kinetic energy will be gained by an α -particle in going from a point at 70 V to another point at 50 V
 (1) 40keV (2) 40eV
 (3) 40MeV (4) 0eV
3. Two capacitors $C_1 = C$ and $C_2 = 3C$ are connected as shown in figure. Initially, key K is open and capacitor C_1 holds charge Q . after closing the key K, the charge on each capacitor at steady state will be

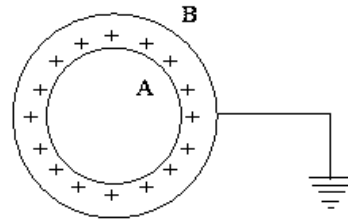


- (1) $\frac{Q}{4}, \frac{Q}{4}$ (2) $\frac{Q}{4}, \frac{3Q}{4}$ (3) $\frac{3Q}{2}, \frac{3Q}{2}$ (4) $\frac{Q}{3}, Q$
4. The energy stored in 4 μF capacitors is



- (1) 8 μJ (2) 12 μJ (3) 16 μJ (4) 20 μJ
5. There is a solid sphere of radius 'R' having uniformly distributed charge. What is the relation between electric field 'E' (inside the sphere) and radius of sphere 'R' is
 (1) $E \propto R^{-2}$ (2) $E \propto R^{-1}$
 (3) $E \propto R^2$ (4) $E \propto \frac{1}{R^3}$

6. A and B are two conducting concentric spherical shells. A is given a charge Q while B is uncharged. If now B is earthed as shown in figure. Then:



- (1) The charge appearing on inner surface of B is $-Q$
 (2) The field between A and B is not zero
 (3) The charge appearing on outer surface of B is zero
 (4) All of these
7. If n drops, each of capacitance C , coalesce to form a single big drop, then the ratio of the energy stored in the big drop to that in each small drop will be
 (1) $n : 1$ (2) $n^{\frac{1}{3}} : 1$ (3) $n^{\frac{5}{3}} : 1$ (4) $n^2 : 1$
8. The electrical potential at a point (x, y, z) is given by $V = -x^2y - xz^3 + 4$
 The electric field $\vec{E}$ at that point is
 (1) $\vec{E} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$
 (2) $\vec{E} = 2xy\hat{i} + (x^2 + y^2)\hat{j} + (3xz - y^2)\hat{k}$
 (3) $\vec{E} = z^3\hat{j} + xyz\hat{i} + 3z^2\hat{k}$
 (4) $\vec{E} = (2xy - z^3)\hat{i} + xy^2\hat{j} + 3z^3x\hat{k}$
9. Just outside a sharp point on a conductor, we will have a large _____ than just outside gradually curving places on the conductor.
 (1) Electric field (2) Potential
 (3) Both (1) and (2) (4) None of these
10. Two thin wire rings each having a radius R are placed at a distance d apart with their axes coinciding. The charges on the two rings are $+q$ and $-q$. The potential difference between the centers of the rings is
 (1) zero
 (2) $\frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{R} - \frac{1}{\sqrt{R^2 - d^2}} \right\}$
 (3) $\frac{qR}{4\pi\epsilon_0 d^2}$
 (4) $\frac{q}{2\pi\epsilon_0} \left\{ \frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right\}$

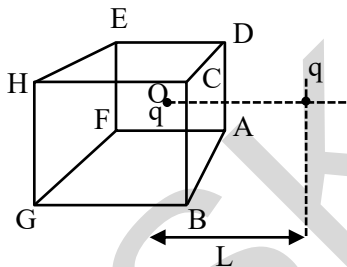
11. Two point charges q and $-q$ are at positions $(0, 0, d)$ and $(0, 0, -d)$ respectively. What is the electric field at $(a, 0, 0)$?

(1) $\frac{2qd}{4\pi\epsilon_0(d^2 + a^2)^{3/2}} \hat{k}$
 (2) $\frac{qd}{4\pi\epsilon_0(d^2 + a^2)^{3/2}} \hat{k}$
 (3) $\frac{-2qd}{4\pi\epsilon_0(d^2 + a^2)^{3/2}} \hat{k}$
 (4) $\frac{-qd}{4\pi\epsilon_0(d^2 + a^2)^{3/2}} \hat{k}$

12. Consider two concentric spherical surface S_1 with radius R and S_2 with radius $2R$ both centered at the origin. There is a charge $+2q$ at the origin and there are no other charges. Compare the flux ϕ_1 through S_1 with the flux ϕ_2 through S_2

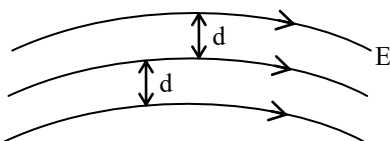
(1) $\phi_1 = 2\phi_2$ (2) $\phi_1 = 4\phi_2$
 (3) $\phi_1 = \phi_2$ (4) $\phi_1 = \frac{\phi_2}{2}$

13. A charge q is placed at the center O of cube of length L . Another same charge q is placed at a distance L from O . Then electric flux through face $ABCD$ is



(1) $\frac{q}{4\pi\epsilon_0 L}$ (2) $\frac{q}{2\pi\epsilon_0 L}$
 (3) $\frac{2q}{2\pi\epsilon_0 L}$ (4) zero

14. For the following electric field lines electric field is –



- (1) Non uniform
 (2) Uniform
 (3) Both uniform & non uniform
 (4) Can't be said

15. A circular ring carries a charge Q , the variation of electric field with distance x measured from center along axis for $x \gg R$ can be given as [$R \rightarrow$ Radius of ring]

(1) $E \propto \frac{1}{x^3}$ (2) $E \propto x$
 (3) $E \propto \frac{1}{x^2}$ (4) $E \propto \frac{1}{x}$

16. The linear charge density of the semicircular ring on both side is same in magnitude. The electric field intensity at O is along

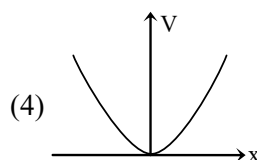
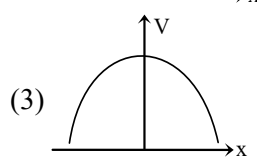
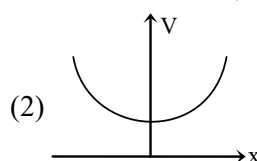
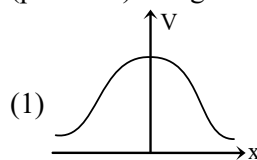


(1) $\hat{i}$ (2) $-\hat{i}$
 (3) $\hat{j}$ (4) $-\hat{j}$

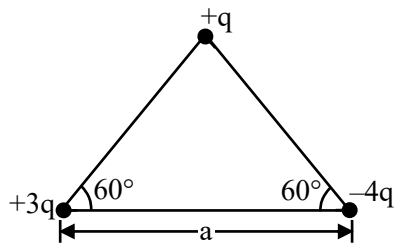
17. An electric dipole is placed at the center of a sphere select the correct option.

- (1) The electric field zero at every point of the sphere
 (2) The flux of the electric field through the sphere is zero
 (3) The electric field is not zero at anywhere on the sphere
 (4) Both (2) and (3)

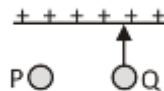
18. Two identical positive charges are placed on the y -axis at $y = -a$ and $y = +a$. The variation of V (potential) along x -axis is shown by graph -



19. Find the magnitude of net dipole moment of the following system.



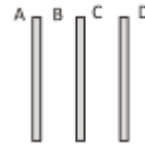
- (1) $qa\sqrt{21}$ (2) $qa\sqrt{13}$
 (3) $4qa$ (4) Zero
20. The unit of electric flux is
- (1) $\frac{N-m^2}{C^2}$
 (2) $\frac{N}{C^2-m^2}$
 (3) V-m
 (4) V-m³
21. In a Millikan-type experiment there are two oil droplets P and Q between the charged horizontal plates, as shown in the diagram. Droplet P is in rest while droplet Q is moving upwards. The polarity of the charges on P and Q is



- | P | Q |
|-------------|---|
| (1) + | + |
| (2) neutral | - |
| (3) - | - |
| (4) + | - |

22. An electric dipole is placed in an electric field generated by a point charge
- (1) The net force on the dipole must be zero.
 (2) The net force on the dipole may be zero.
 (3) The torque on the dipole due to the field must be zero.
 (4) The torque on the dipole due to the field may be zero.
23. To form a composite $16\mu\text{F}$, 1000V capacitor from a supply of identical capacitors marked $8\mu\text{F}$, 250V , we require a minimum number of n capacitors, where n is:
- (1) 2 (2) 8 (3) 16 (4) 32

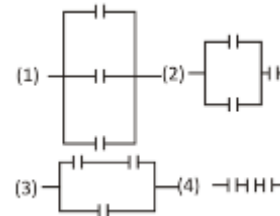
24. The figure below shows an edge-on view of three extremely large, parallel insulating sheets positively charged with uniform and identical surface charge densities. In which of the regions shown is the electric field maximum?



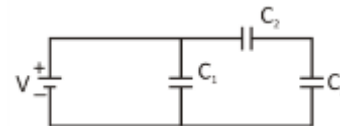
- (1) B
 (2) C
 (3) A and D
 (4) The electric field is the same in all four regions of space.
25. A charge Q_0 is placed at centre of spherical cavity of a conductor and another charge Q is placed outside the conductor. The correct statement is:



- (1) Force on Q_0 and Q are equal and opposite
 (2) Force on Q_0 is zero and force on Q is non-zero
 (3) Force on Q and Q_0 is zero
 (4) Force on Q_0 is non-zero and force on Q is zero
26. In four options below all the four circuits are arranged in order of equivalent capacitance. Select the correct order. Assume all capacitors are of equal capacitance.



- (1) $C_1 > C_2 > C_3 > C_4$
 (2) $C_1 > C_3 > C_2 > C_4$
 (3) $C_1 < C_2 < C_3 < C_4$
 (4) $C_1 < C_3 < C_2 < C_4$
27. Three capacitors C_1 , C_2 and C_3 are connected to a battery as shown in the figure. The three capacitors have equal capacitance. Which capacitor stores the most energy?



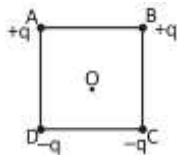
- (1) C_2 or C_3 as they store the same amount of energy
 (2) C_2
 (3) C_1
 (4) All three capacitors store the same amount of energy

28. The work done in bringing a 20C charge from point A to point B for distance 0.2m is 2J. The potential difference between the two points will be:
 (1) 0.2V (2) 8V
 (3) 0.1V (4) 0.4V

29. Electric potential at any point in a region is given as $V = -5x + 3y + \sqrt{15}z$.
 In this region the magnitude of the electric field is:

- (1) $3\sqrt{2}$ (2) $4\sqrt{2}$
 (3) $5\sqrt{2}$ (4) 7

30. Let $\vec{E}$ be the E-field and V the potential at of centre of square having charges on the vertices as shown in figure. If the charges on A and B are interchanged with those on D and C respectively, then.

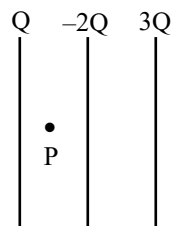


- (1) E change V remains same
 (2) E remains same. V changes
 (3) Both changed
 (4) Both remains same.

31. A thin iron ring of radius r has an electric charge q. What will be the force on a point charge q_0 placed at the centre of the ring.

- (1) zero (2) $\frac{Kqq_0}{r^2}$
 (3) $\frac{2Kqq_0}{r^2}$ (4) $\frac{Kq}{r^2}$

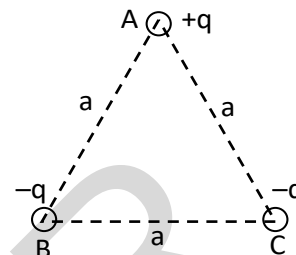
32. Three large identical conducting sheets placed parallel to each other at finite distance contains charges Q, -2Q and 3Q respectively. If surface area of sheet is A, then magnitude of electric field at point P is -



- (1) Zero (2) $\frac{2Q}{A \epsilon_0}$
 (3) $\frac{Q}{A \epsilon_0}$ (4) $\frac{4Q}{A \epsilon_0}$

33. Two point charges when kept in air at a separation 1m exert 810N repulsive force. Now if the same pair is kept in water at same separation force between them will become
 (1) 81 N (2) 270 N (3) 10 N (4) zero

34. ABC is an equilateral triangle of side a. Three point charges are placed at the three vertices as shown in the figure. Resultant force on the charge at vertex B will be



- (1) $\sqrt{2} \frac{kq^2}{a^2}$ (2) $\sqrt{3} \frac{kq^2}{a^2}$
 (3) $\frac{kq^2}{a^2}$ (4) $\frac{kq^2}{2a^2}$

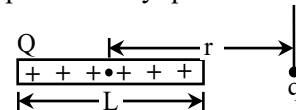
35. A particle having mass 5 mg and charge $6\mu\text{C}$ is released in uniform electric field of magnitude $1 \times 10^2 \text{ NC}^{-1}$. Acceleration of the particle in ms^{-2} will be

- (1) 1.2×10^5 (2) $\frac{5}{6} \times 10^5$
 (3) 1.2×10^2 (4) $\frac{5}{6} \times 10^2$

36. At a point very close to the surface of a charged conductor electric field makes an angle θ with the surface of the conductor. θ equal to

- (1) 90° (2) 0° (3) 60° (4) 45°

37. A charge Q is distributed over a line of length L. Another point charge q is placed at a distance r from the centre of the line distribution. Then the force experienced by q is-



- (1) $\frac{qQ}{4\pi \epsilon_0 (r^2 - L^2)}$ (2) $\frac{qQ}{4\pi \epsilon_0 (r^2 - (L/2)^2)}$
 (3) $\frac{qQ}{4\pi \epsilon_0 r^2}$ (4) $\frac{qQL}{4\pi \epsilon_0 r^3}$

38. At an axial position at a distance r from a small electric dipole, field produced by the dipole has magnitude E. At another axial position at a distance 2r from the dipole it will produce field having magnitude

- (1) 4E (2) $\frac{E}{4}$ (3) 8E (4) $\frac{E}{8}$

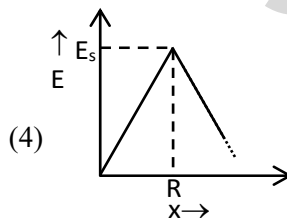
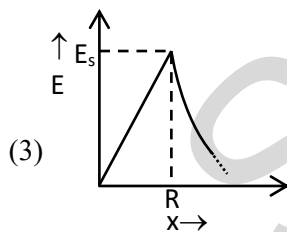
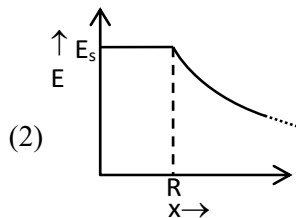
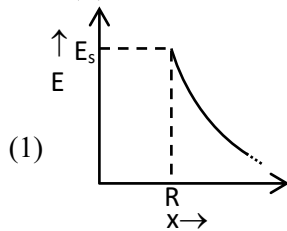
39. A solid copper sphere of radius 20 cm is positively charged. At a distance 50 cm from its centre electric field is found to be 10^3 NC^{-1} . Electric field at a distance 10 cm from its centre will be

(1) $25 \times 10^3 \text{ NC}^{-1}$ (2) $5 \times 10^3 \text{ NC}^{-1}$
 (3) $9 \times 10^3 \text{ NC}^{-1}$ (4) Zero

40. Two point charges $16 \mu\text{C}$ and $-4 \mu\text{C}$ are located at positions $X = 0$ and $X = 4 \text{ cm}$ on the x-axis respectively. Resultant electric field will be zero at

(1) $X = -8 \text{ cm}$ (2) $X = 2 \text{ cm}$
 (3) $X = 8 \text{ cm}$ (4) $X = 12 \text{ cm}$

41. A non-conducting solid sphere of radius R is uniformly charged. Which of the following graphs correctly shows the variation of electric field (E) w.r.t. distance from its centre (x)



42. A long straight wire is uniformly charged with linear charge density λ . Electric field at a perpendicular distance r from the wire is proportional to

(1) $E \propto \lambda r$
 (2) $E \propto \frac{r}{\lambda}$
 (3) $E \propto \frac{\lambda}{r}$
 (4) $E \propto \frac{\sqrt{\lambda}}{r}$

43. In a region uniform electric field is given by $\vec{E} = \alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}$. Electric flux through a square of side l , lying on zx plane can be given by

(1) $\phi = l^2 \sqrt{\alpha^2 + \gamma^2}$
 (2) $\phi = l^2 \beta$
 (3) $\phi = l^2 \sqrt{\alpha^2 + \beta^2 + \gamma^2}$
 (4) $\phi = 0$

44. A square has side length a . A point charge q is placed at a height $h = a/2$ above centre of the square. Electric flux associated with the square is

(1) $\phi = \frac{q}{2\epsilon_0}$
 (2) $\phi = \frac{q}{\epsilon_0}$
 (3) $\phi = \frac{q}{6\epsilon_0}$
 (4) $\phi = \frac{q}{8\epsilon_0}$

45. A large non-conducting sheet of negligible thickness is uniformly charged. Electric field (E) at a perpendicular distance (r) w.r.t. the sheet varies as

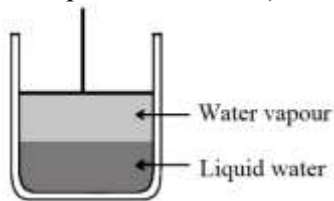
(1) $E \propto r^0$
 (2) $E \propto r^{-1}$
 (3) $E \propto r^{-2}$
 (4) $E \propto r^{-3}$

CHEMISTRY

46. When mercuric iodide is added to the aqueous solution of KI then the:
 (1) Freezing point is raised
 (2) Freezing point is lowered
 (3) Freezing point does not change
 (4) boiling point does not change
47. A 1.0g sample of $\text{Co}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3\text{Cl}_3$ is dissolved in 25.0g of water and the freezing point of the solution is -0.87°C . How many ions are produced per mole of compound? the k_f of water is 1.86°C/molal
 (1) 2 (2) 3 (3) 4 (4) 5
48. The freezing point of a 4% aqueous solution of 'A' is equal to the freezing point of 10% aqueous solution of 'B'. If the molecular weight of 'A' is 60, then the molecular weight of 'B' will be:
 (1) 160 (2) 90 (3) 45 (4) 180
49. At 300K, 40 mL of $\text{O}_{3(g)}$ dissolves in 100 g of water at 1.0 atm. What mass of ozone dissolved in 400g of water at a pressure of 4.0 atm at 300 K?
 (1) 0.1 g (2) 1.2 g
 (3) 0.48 g (4) 4.8 g
50. The charge required for the oxidation of two moles Mn_3O_4 into MnO_4^{2-} in presence of alkaline medium is:
 (1) 5F (2) 10F
 (3) 20F (4) None of these
51. How much time is required for complete decomposition of 4 moles of water using 4 ampere?
 (1) 3.86×10^5 sec (2) 1.93×10^5 sec
 (3) 96500 sec (4) 48250 sec
52. For a reaction $\text{A}_{(s)} + 2\text{B}^+ \rightleftharpoons \text{A}^{2+} + 2\text{B}_{(s)}$ K_c has been found to be 10^{12} . The E°_{cell} is:
 (1) 0.354 V (2) 0.708 V
 (3) 0.0098 V (4) 1.36 V
53. Which of the following displacement does not occur.
 (1) $\text{Zn} + 2\text{H}^+ \longrightarrow \text{Zn}^{2+} + \text{H}_2 \uparrow$
 (2) $\text{Fe} + 2\text{Ag}^+ \longrightarrow \text{Fe}^{2+} + \text{Ag} \downarrow$
 (3) $\text{Cu} + \text{Fe}^{2+} \longrightarrow \text{Cu}^{2+} + \text{Pb} \downarrow$
 (4) $\text{Zn} + \text{Pb}^{2+} \longrightarrow \text{Zn}^{2+} + \text{Pb} \downarrow$
54. A solution of two liquids boils at a temperature more than the boiling point of either of them. Hence, the binary solution shows:
 (1) Negative deviation from Raoult's law
 (2) Positive deviation from Raoult's law
 (3) No deviation from Raoult's law
 (4) Positive or negative deviation from Raoult's law depending upon the composition.
55. The Van't Hoff factor of 0.005 M aqueous solution of KCl is 1.95. The degree of ionisation of KCl is:
 (1) 0.95 (2) 0.97 (3) 0.94 (4) 0.96
56. Low concentration of oxygen in the blood and tissues of people living at high altitude is due to:
 (1) Low temperature
 (2) Low atmospheric pressure
 (3) High atmospheric pressure
 (4) Both low temperature and high atmospheric pressure
57. At a given temperature, osmotic pressure of a concentrated solution of a substance:
 (1) Is higher than that of a dilute solution
 (2) Is lower than that of a dilute solution
 (3) Is same as that of a dilute solution
 (4) Cannot be compared with osmotic pressure of dilute solution
58. A living cell contains a solution which is isotonic with 0.2 M glucose solution. What osmotic pressure develops when the cell is placed in 0.05 M BaCl_2 solution at 300 K?
 (1) 1.23 atm (2) 6.15 atm
 (3) 3.69 atm (4) None of these
59. The vapour pressure of a solvent decreases by 10 mm of Hg than a non-volatile solute was added is 0.2. What should be the mole fraction of the solvent if the decrease in vapour pressure is to be 20 mm of Hg.
 (1) 0.6 (2) 0.4
 (3) 0.2 (4) 1
60. The electrode potentials for $\text{Cu}^{2+}(\text{aq}) + \text{e}^- \rightarrow \text{Cu}^+(\text{aq})$ and $\text{Cu}^+(\text{aq}) + \text{e}^- \rightarrow \text{Cu}(\text{aq})$ are 0.15 V and + 0.50 V respectively. The value of $E^\circ_{\text{Cu}^{2+}/\text{Cu}}$ will be
 (1) 0.150 V (2) 0.500 V
 (3) 0.325 V (4) 0.650 V

61. Molar conductance of KCl increases slowly with decrease in concentration because of
 (1) Increase in degree of ionisation
 (2) Increase in total number of current carrying species
 (3) Weakening of interionic attractions and increase in ionic mobilities
 (4) Increase in hydration of ions
62. Equivalent conductance of 1M of CH_3COOH is $10 \text{ ohm}^{-1}\text{cm}^2\text{eq}^{-1}$ and that at infinite dilution is $200 \text{ ohm}^{-1}\text{cm}^2\text{eq}^{-1}$. Hence the % ionization of CH_3COOH is
 (1) 5% (2) 2% (3) 4% (4) 1%
63. The emf of the cell
 $\text{Ni}/\text{Ni}^{+2} (1.0 \text{ M}) \parallel \text{Au}^{+3} (0.1 \text{ M})/\text{Au}$
 $[\text{E}^\circ \text{ for } \text{Ni}^{+2}/\text{Ni} = -0.25\text{V}, \text{E}^\circ \text{ for } \text{Au}^{+3}/\text{Au} = 1.50 \text{ V}]$
 is given as :-
 (1) 1.25 V (2) -1.75 V
 (3) 1.78 V (4) 1.73V
64. An aqueous solution containing 1M each of Au^{3+} , Cu^{2+} , Ag^+ , Li^+ is being electrolysed by using inert electrodes. The value of standard potentials are:
 $\text{E}^\circ_{\text{Ag}^+/\text{Ag}} = 0.80\text{V}$, $\text{E}^\circ_{\text{Cu}^+/\text{Cu}} = 0.34\text{V}$ and
 $\text{E}^\circ_{\text{Au}^{3+}/\text{Au}} = 1.50\text{V}$, $\text{E}^\circ_{\text{Li}^+/\text{Li}} = -3.03\text{V}$
 With increasing voltage, the sequence of deposition of metals on the cathode will be:
 (1) Li, Cu, Ag, Au
 (2) Cu, Ag, Au
 (3) Au, Ag, Cu
 (4) Au, Ag, Cu, Li
65. What is the mole fraction of glucose in 10% w/w glucose solution?
 (1) 0.01 (2) 0.02
 (3) 0.03 (4) 0.04
66. The amount of solute (molar mass 60 g mol^{-1}) that must be added to 180 g of water so that the vapour pressure of water is lowered by 10% is:
 (1) 30 g (2) 60 g (3) 120 g (4) 12 g
67. Which of the following will show a negative deviation from Raoult's law?
 (1) Acetone - benzene
 (2) Acetone - ethanol
 (3) Benzene - methanol
 (4) Acetone - chloroform
68. Match the terms given in Column I with the units given in Column II.
- | Column-I | | Column-II | |
|----------|--------------------------|-----------|-------------------------------|
| (a) | Λ_m | (p) | Scm^{-1} |
| (b) | E_{cell} | (q) | m^{-1} |
| (c) | k | (r) | $\text{Scm}^2\text{mol}^{-1}$ |
| (d) | G^* | (s) | V |
- (1) $a \rightarrow r, b \rightarrow s, c \rightarrow p, d \rightarrow q$
 (2) $a \rightarrow p, b \rightarrow q, c \rightarrow r, d \rightarrow s$
 (3) $a \rightarrow q, b \rightarrow p, c \rightarrow r, d \rightarrow s$
 (4) $a \rightarrow r, b \rightarrow p, c \rightarrow s, d \rightarrow q$
69. **Assertion:** E_{cell} should have a positive value for the cell to function.
Reason: $\text{E}_{\text{cathode}} < \text{E}_{\text{anode}}$
 (1) Both Assertion and reason are true and the reason is the correct explanation of assertion.
 (2) Both assertion and reason are true and the reason is not correct explanation for assertion
 (3) Assertion is true but reason is false.
 (4) Assertion is false but reason is true
70. Consider the following solutions
 I. 1 M aqueous solution of sucrose
 II. 0.5 M aqueous solution of KCl
 III. 2M aqueous solution of benzoic acid
 IV. 0.25M aqueous solution of $(\text{NH}_4)_3\text{PO}_4$
 Select the correct statement.
 (1) All are isotonic
 (2) III is hypertonic with respect to I, II and IV
 (3) IV is hypotonic with respect to I, II, III
 (4) All are hypotonic
71. Blood is isotonic with.
 (1) 0.9% (w/v) - NaCl solution
 (2) 0.9 m - NaCl solution
 (3) 0.9 M - NaCl solution
 (4) 9.0% (w/v) - NaCl solution
72. **Assertion :** If a non-volatile solute is added to a solvent then the vapour pressure of solution at a given temperature is found to be lower than vapour pressure of pure solvent
Reason : On addition of non-volatile solute, the number of solvent molecules escaping from the surface is correspondingly reduced.
 (1) Both assertion and Reason are correct and Reason is correct explanation of the Assertion.
 (2) Both assertion and Reason are correct but Reason is not correct explanation of the Assertion.
 (3) Assertion is correct but Reason is incorrect
 (4) Assertion is incorrect but reason is correct

73. The vapour pressure of water at 20°C is 17.54 mm of Hg. What will be the vapour pressure of the water in the apparatus shown after the piston is lowered, decreasing the volume of the gas above the liquid to one half of its initial volume (assume temperature constant).



- (1) 8.77 mm of Hg
(2) 17.54 mm of Hg
(3) 35.08 mm of Hg
(4) between 8.77 and 17.54 mm of Hg
74. If 1 mole A and 4 mole B on mixing have total pressure is 75 torr, vapour pressure of pure A is 200 torr and vapour pressure of pure B is 50 torr. Which of the following is correct :
- (1) $\Delta S_{\text{mix}} = +\text{ve}$ (2) $\Delta H_{\text{mix}} = +\text{ve}$
(3) $\Delta G_{\text{mix}} = +\text{ve}$ (4) $\Delta V_{\text{mix}} = +\text{ve}$

75. Match the columns:

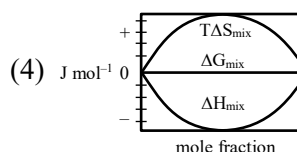
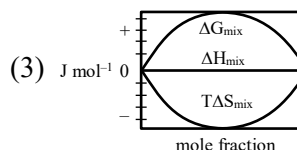
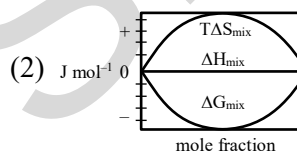
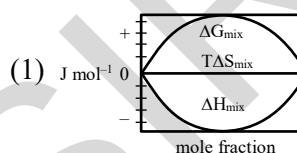
Column-I		Column-II	
(A)	Na-Hg Amalgam	(p)	gas in solid
(B)	H_2 in Pd	(q)	gas in liquid
(C)	Camphor in nitrogen gas	(r)	liquid in solid
(D)	Oxygen dissolved in water	(s)	solid in gas

- (1) A – (q), B – (s), C – (r), D – (p)
(2) A – (r), B – (p), C – (q), D – (s)
(3) A – (r), B – (p), C – (s), D – (q)
(4) A – (s), B – (q), C – (p), D – (r)
76. **Assertion :** If on mixing the two liquids, the solution becomes hot, it implies that it shows negative deviation from Raoult's law.
Reason : Solution which show negative deviation are accompanied by decrease in volume.
- (1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true but reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.
77. What will be molality of CH_4 in benzene at 25°C using Henry's law. When partial pressure of CH_4 is 760 mm and K_H is 4.27×10^5 mm Hg ?
- (1) 0.023 m (2) 9.5 m
(3) 6.5 m (4) 28.3 m

78. According to Henry's law the partial pressure of the gas in vapour phase (p) is proportional to the mole fraction of the gas (x) in the solution'. For different gases the correct statement about Henry's constant is

- (1) Higher the value of K_H at a given pressure, higher is the solubility of the gas
(2) Higher the value of K_H at a given pressure, lower is the solubility of the gas
(3) K_H is not a function of nature of gas
(4) K_H value for all gases is same at a given pressure

79. Which of the following represents correctly the changes in thermodynamic properties during the formation of 1 mole of an ideal binary solution :

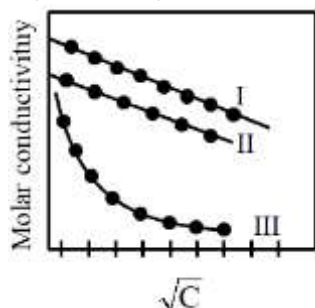


80. **Assertion:** The molar conductance of weak electrolytes is low as compared to that of strong electrolytes at moderate concentrations.

Reason: Weak electrolytes at moderate concentration dissociates to a much greater extent as compared to strong electrolytes.

- (1) Both assertion and reason are correct but reason is not the correct explanation of assertion.
(2) Assertion is correct but reason is not correct.
(3) Assertion is not correct but reason is correct.
(4) Both assertion and reason are correct and reason is the correct explanation of assertion.

81. A graph was plotted between molar conductivity of various electrolytes (NaCl, HCl and NH_4OH) and $\sqrt{C}$ (in mol L^{-1}). Correct setting :



- (1) I(NaCl), II(HCl), III(NH_4OH)
 (2) I(HCl), II(NaCl), III(NH_4OH)
 (3) I(NH_4OH), II(NaCl), III(HCl)
 (4) I(NH_4OH), II(HCl), III(NaCl)
82. The weight of silver (at wt. = 108) displaced by a quantity of electricity which displaces 5600 mL of O_2 at STP will be :-
- (1) 5.4 g (2) 10.8 g
 (3) 54.0 g (4) 108.0 g
83. Resistance of 0.01 N CaCl_2 solution is 210 ohm at 298 K with cell constant 0.84 cm^{-1} . Match the column.

	Column-I		Column-II
(i)	Conductance (ohm^{-1})	(a)	4×10^{-3}
(ii)	Specific conductance ($\text{ohm}^{-1} \text{ cm}^{-1}$)	(b)	400
(iii)	Molar conductance ($\text{ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$)	(c)	800
(iv)	Equivalent conductance ($\text{ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$)	(d)	$\frac{1}{210}$

- (1) (i) $\rightarrow$ a, (ii) $\rightarrow$ b, (iii) $\rightarrow$ c, (iv) $\rightarrow$ d
 (2) (i) $\rightarrow$ d, (ii) $\rightarrow$ a, (iii) $\rightarrow$ c, (iv) $\rightarrow$ b
 (3) (i) $\rightarrow$ d, (ii) $\rightarrow$ a, (iii) $\rightarrow$ b, (iv) $\rightarrow$ c
 (4) (i) $\rightarrow$ d, (ii) $\rightarrow$ c, (iii) $\rightarrow$ a, (iv) $\rightarrow$ b
84. Consider the following statements :-
- A battery is an arrangement of electrolytic cells.
 - A paste of HgO and carbon is used as an cathode in mercury cell.
 - Lead storage cell has longer life than Nickel-Cadmium cell.
 - In dry cell the space between the electrodes is filled by a moist paste of ammonium chloride and zinc chloride.
- Correct statements are
- (1) a and b only (2) b and d only
 (3) a and c only (4) c and d only

85. Two half reactions with E^0 values are given below
 (i) $\text{Fe}^{3+} + e^- \rightarrow \text{Fe}^{2+}$; $E^0 = +0.77 \text{ V}$
 (ii) $\text{Fe}^{2+} + 2e^- \rightarrow \text{Fe}$; $E^0 = -0.44 \text{ V}$
 The standard potential for the cell reaction $2\text{Fe}^{3+} + \text{Fe} \rightarrow 3\text{Fe}^{2+}$
- (1) +0.33 V (2) +1.10 V
 (3) +1.21 V (4) +1.43 V
86. Which of the following statements about the lead storage battery is incorrect ?
- The anode consists of lead and the cathode of lead dioxide.
 - The electrolyte is an aqueous solution of sulphuric acid.
 - It is a type of secondary battery.
 - As the battery runs down lead sulphate is formed on the anode and lead sulphite is formed on the cathode.
87. During electrolysis of fused calcium hydride CaH_2 , the hydrogen will produce at
- cathode
 - anode
 - Hydrogen will not liberated at all
 - H_2 produced will react with O_2 give H_2O
88. What is the value of ΔG^0 for the reaction? $2\text{Al(s)} + 3\text{Cu}^{2+}(\text{aq}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{Cu(s)}$
 $E^0 = 2.02 \text{ V}$
- (1) -1170 kJ (2) -585 kJ
 (3) -390 kJ (4) -195 kJ
89. In leclanche cell oxidation state of Mn change from
- (1) +7 to +6 (2) +7 to +2
 (3) +4 to +3 (4) +7 to +4
90. Match the column-I with column-II for Daniel cell and choose the correct option.

	Column-I		Column-II
(A)	Flow of electrons from Cu to Zn and current flows from Zn to Cu	(p)	$E_{\text{ext}} = 1.1 \text{ V}$
(B)	No flow of electrons or current	(q)	$E_{\text{ext}} < 1.1 \text{ V}$
(C)	Zn dissolves at anode and copper deposits at cathode	(r)	$E_{\text{ext}} > 1.1 \text{ V}$

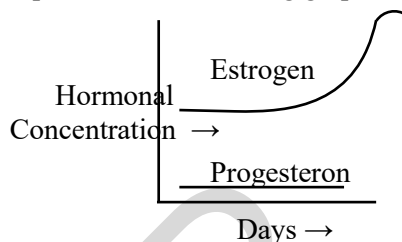
- (1) (A) - (q), (B) - (p), (C) - (r)
 (2) (A) - (r), (B) - (p), (C) - (q)
 (3) (A) - (p), (B) - (r), (C) - (q)
 (4) (A) - (r), (B) - (q), (C) - (p)

BIOLOGY

91. Regarding test tube baby programme, select the correct sequential steps.
 (A) Completion of further development of zygote/embryo
 (B) Formation of zygote under simulated conditions in the laboratory
 (C) Collection of ova from the wife/donor(female) and sperms from the husband/donor (male)
 (D) Transfer of zygote/embryo into the fallopian tube/ uterus
 Choose the correct answer from the options given below:
 (1) A, D, C, B (2) C, B, D, A
 (3) B, A, D, C (4) D, A, C, B
92. Middle layers of microsporangia are present between
 (1) Epidermis and endothecium
 (2) Endothecium and tapetum
 (3) Epidermis and tapetum
 (4) Inside the tapetum
93. Which is the correct sequence of microsporogenesis?
 (1) Microspore tetrad → Pollen mother cell → Sporogenous tissue
 (2) Sporogenous tissue → microspore tetrad → Pollen mother cell
 (3) Pollen mother cell → microspore tetrad → Sporogenous tissue
 (4) Sporogenous tissue → Pollen mother cell → microspore tetrad
94. Part of exine where sporopollenin is not present are called
 (1) Germ pore (2) Micropyle
 (3) Generative part (4) Chalazal end
95. Funicle is
 (1) Stalk of the ovule
 (2) Where body of the ovules fuse with hilum
 (3) Protective covering of body
 (4) Present at chalazal end
96. Select the mismatched pairs
 (1) Microsporangium – pollen sac
 (2) Megasporangium – Ovule
 (3) Pollen grain – female gametes
 (4) Embryo sac – Female gametophyte
97. In angiosperm, two non-motile male gametes are produced by
 (1) Generative cell
 (2) Microspore mother cell
 (3) Vegetative cell
 (4) Tube cell

98. Transfer of pollen grains from the anther to the stigma of another flower of same plant is called
 (1) Xenogamy (2) Autogamy
 (3) Geitonogamy (4) Allogamy

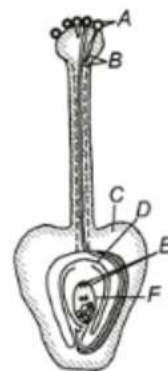
99. The level of estrogen and progesterone hormones in a particular phase of menstrual cycle are represented in following graph.



Which of the following is true about this particular phase of menstrual cycle?

- (1) Transformation of remaining part of Graafian follicle as corpus luteum
 (2) Completion of the second meiosis division of the oocyte
 (3) Breakdown of endometrial lining of the uterus
 (4) Gradual increase in the secretion of gonadotropins
100. What is correct when chromosome number in leaf cells of an angiosperm is 22?
 (1) 44 in stem cells (2) 44 in embryo
 (3) 11 in gametes (4) 22 in gametes
101. Outbreeding devices tend to promote
 (1) Autogamy (2) Self-pollination
 (3) Xenogamy (4) Inbreeding depression
102. Function of filiform apparatus is to
 (1) Recognize the suitable pollen at stigma
 (2) Stimulate division of generative cell
 (3) Produce nectar
 (4) Guide the entry of pollen tube into ovule.
103. Given below are two statements:
Statement-I : Exine of pollen grain is made up of sporopollenin.
Statement-II : Intine of pollen grain is made up of pectocellulose.
 In the light of the above statements, choose the most appropriate answer from the options given below :
 (1) Both statement-I and statement-II are correct.
 (2) Statement-I is correct but statement-II is incorrect.
 (3) Statement-I is incorrect and statement-II is correct.
 (4) Both statement-I and statement-II are incorrect.

104. In most of angiosperms, pollen grains are shed in
 (1) 2 celled stage (2) 2-3 celled stage
 (3) 3-celled stage (4) 4-celled stage
105. If stem has $2n = 10$ number of chromosomes then find out
 A- number of chromosomes in endosperm.
 B- number of chromosomes in egg cell.
 C- number of chromosomes in each polar nucleus, respectively.
 (1) 15, 15, 20 (2) 10, 15, 20
 (3) 15, 5, 5 (4) 10, 5, 15
106. A typically angiosperm embryo sac is
 (1) 8-nucleate, 7-celled
 (2) 7-celled, 7-nucleate
 (3) 7-nucleate, 8-celled
 (4) 8-nucleate, 8-celled
107. Medical assistance and care to people is required in reproduction related problems like:
 (a) Pregnancy (b) Delivery
 (c) STDs (d) Abortions
 (e) Contraception (f) Menstrual problems
 (g) Infertility
 (1) a, c, e and f
 (2) b, d and g
 (3) a, b, c, d, d, e and g
 (4) a, b, c, d, e, f and g
108. Which of the following is short lived in micro sporangium
 (1) Epidermis (2) Middle layer
 (3) Endothecium (4) Tapetum
109. Wind pollination is found in
 (1) Fig (2) Grasses
 (3) *Yucca* (4) *Vallisneria*
110. Read the following statements and choose the correct statements.
 I. Plants use two abiotic (wind, water) and one biotic (animals) agents to achieve pollination.
 II. Cleistogamous flower is present in *Commelina*
 III. Tallest flower are of *Amorphophallus*.
 IV. In monoecious condition, both male and female flowers are borne on same plant, an example of such plant is Maize.
 (1) I and II (2) II and III
 (3) III and IV (4) I, II, III and IV
111. When male gametes are non-motile but they are carried by pollen tube called :
 (1) Porogamy (2) Zooidiogamy
 (3) Siphonogamy (4) Syngamy
112. Pollen tablets & Syrups are nutritious and used by:
 (1) Horses of race and Athletes
 (2) Pigs and Buffalo
 (3) Athletes and Monkeys
 (4) None of these
113. Select wrong statement for cleistogamy of flower in plant.
 (1) Cleistogamous flower lack requirement of pollinators
 (2) It is a adaptation of autogamy
 (3) It is performed in *Oxalis* and *Viola*
 (4) Their flower are coloured contain aroma and nectar
114. The egg apparatus of angiosperm comprises
 (1) An egg cell and two antipodals
 (2) An egg cell and two synergids
 (3) An egg cell and two polar nuclei
 (4) An egg cell and the central cell
115. Pollen allergy is mainly caused by
 (1) Rice and Wheat
 (2) Rosaceae, Leguminosae
 (3) *Parthenium* or carrot grass
 (4) *Agave*
116. Synergid, zygote and primary endosperm nucleus respectively are
 (1) haploid, diploid and triploid
 (2) haploid, diploid and diploid
 (3) diploid, diploid and triploid
 (4) triploid, haploid and diploid
117. The phenomenon observed in some plants wherein parts of the sexual apparatus is used for forming embryos without fertilization called:
 (1) Parthenocarpy
 (2) Adventive embryony
 (3) Vegetative propagation
 (4) Sexual reproduction
118. Identify A to F in the given diagram.



- (1) A-Pollen tube, B-Ovary, C-Ovule, D-Antipodal cell, E-Pollen grain, F-Polar nuclei
 (2) A-polar nuclei, B-Antipodal cell, C-Ovule, D-Ovary, E-Pollen tube, F-Pollen grain
 (3) A-Pollen grain, B-Pollen tube, C-Ovary, D-Ovule, E-Antipodal cell, F-Polar nuclei
 (4) A-Antipodal cell, B-Ovule, C-Ovary, D-Polar nuclei, E-Pollen grain, F-Pollen tube
- 119.** Which of the following is not a part of female gametophyte of angiosperms
 (1) Antipodals (2) Polar nuclei
 (3) Synergids (4) Prothallial cell
- 120.** The central cell after triple fusion becomes the:
 (1) Zygote
 (2) Cell of embryo
 (3) Primary endosperm cell
 (4) Suspensor cell
- 121.** Perisperm is:
 (1) Degenerate part of synergids
 (2) Peripheral part of endosperm
 (3) Degenerate part of secondary nucleus
 (4) Remains of nucellus
- 122.** Autogamy occurs in:
 (1) between two flower of same plant
 (2) In same flower
 (3) between two flower of different plant of same species
 (4) none of these
- 123.** Given below are two statements :
Statement-I : To some extent viability of pollen grains depends upon prevailing temperature and humidity.
Statement-II : Viability of pollen grains is lost in few minutes in members of family solanaceae.
 In the light of the above statements, choose the most appropriate answer from the options given below :
 (1) Both statement-I and statement-II are correct.
 (2) Statement-I is correct but statement-II is incorrect.
 (3) Statement-I is incorrect and statement-II is correct.
 (4) Both statement-I and statement-II are incorrect.
- 124.** Consider the following statements given below and choose the option. Which include correct statements only?
 (A) Contraception pills include administration of progesterone or progesterone-estrogen combinations.
 (B) Medical termination of pregnancy (MTP) is safe during third trimester.
 (C) First seven days and last seven days of menstrual cycle is called risk period.
 (D) Intrauterine devices like Cu-T are effective contraceptives.
 (1) B & D (2) A & D
 (3) B & C (4) A, B & D
- 125.** Given below are four statement (a-d) regarding assisted reproductive technologies -
 (a) ZIFT – The zygote or early embryo (with upto 8 blastomere) transferred into the fallopian tube
 (b) ICSI – A sperm is directly injected into the fallopian tube to form an embryo in the laboratory
 (c) IUI – The semen collected either from the husband or healthy donor is artificially introduced into the uterus.
 (d) GIFT – Transfer of zygote collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development
 Which two of the above statements are correct?
 (1) a and d (2) b and c
 (3) a and c (4) b and d
- 126.** Which one of the following statement is incorrect about pollination?
 (1) Anemophily is by wind and occurs in grasses and date palm
 (2) Hydrophily is by water and occurs in *Zostera*, *Vallisneria* and *Ceratophyllum*
 (3) Entomophily is by insects and occurs in rose, jasmine, *Salvia*, etc.
 (4) Xenogamy increases inbreeding depression
- 127.** Apomixis is useful in plant as it :
 (1) Prevents segregation of characters in hybrid progeny
 (2) Helps in production of pure line progeny by sexual reproduction
 (3) It requires more than one parental plants for reproduction process
 (4) None of these

128. Study the following and find correct statements(s).
- Pollination by water quite in flowering plants and is limited to about 50 genera, mostly Monocotyledons.
 - Water is a regular mode transport for the male gametes among higher plant groups.
 - Distribution of bryophytes andpteridophytes are limited because of the need of water for transport of male gametes and fertilization.
- (1) Only I
 - (2) II and III
 - (3) Only III
 - (4) Only II

129. Given below are two statements (for angiospermic plants) :

Statement-I : A mature male gametophyte possess two male gametes and one vegetative cell.

Statement-II : A mature female gametophyte is 7-celled and 8 nucleated in typical angiosperm. In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both statement-I and statement-II are correct.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect and statement-II is correct.
- (4) Both statement-I and statement-II are incorrect.

130. Given below are two statements ; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Geitonogamy is genetically same as autogamy.

Reason (R) : In geitonogamy transfer of pollen grains occur from anther to stigma of another flower of same plant.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) & (R) are correct but (R) is the not correct explanation of (A).
- (2) (A) is correct but (R) is incorrect.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

131. Match the Column-I and Column-II:

Column-I		Column-II	
A.	Parthenocarpic fruit	(i)	Apomixis
B.	Grasses	(ii)	Excavated from arctic Tundra
C.	Seeds of <i>Lupinus articus</i>	(iii)	2000 years old viable seeds
D.	Date palm	(iv)	Banana

- (1) A-i, B-iv, C-iii, D-ii
- (2) A-iv, B-i, C-ii, D-iii
- (3) A-i, B-ii, C-iii, D-iv
- (4) A-iv, B-i, C-iii, D-ii

132. Identify the correct set of statements for angiosperms.

- Maturation of ovary into fruit is a post fertilization event.
- Nuclellar adventive embryony is found in citrus and mango varieties.
- Free nuclear divisions occur during embryogenesis in angiosperms.
- Filliform apparatus secretes chemicals, which attract pollen tube toward synergids.
- Development of female gametophyte in a typical angiosperm is monosporic.

Choose the correct answer from the options given below :

- (1) a, c and e only
- (2) b and c only
- (3) a, b, d and e only
- (4) b, c, d and e only

133. The corn cob having the tassels which are nothing but:

- (1) The stigma and style
- (2) Only Stigma
- (3) Only style
- (4) Complete pistil

134. **Assertion(A):** Megaspore mother cell undergoes meiosis to produce four megaspores.

Reason(R): Megaspore mother cell and megaspore both are haploid.

- (1) If both A and R are true and R is correct explanation of A
- (2) If both A and R are true, but R is not the correct explanation of A
- (3) If A is true, but R is false
- (4) If A is false, but R is true

135. **Assertion(A):** Some fruits are seedless or contain non-viable seeds.

Reason(R): They are produced without fertilization.

- (1) If both A and R are true and R is correct explanation of A
- (2) If both A and R are true, but R is not the correct explanation of A
- (3) If A is true, but R is false
- (4) If A is false, but R is true

136. Head of sperm contains

- (1) Nucleus having diploid set of chromosomes
- (2) Nucleus having haploid set of chromosomes
- (3) Nucleus and mitochondria
- (4) Mitochondria and acrosome

137. In human, the testes occur in scrotal sacs, outside the abdomen because of the :-

- (1) Presence of urinary bladder
- (2) Presence of rectum
- (3) Long vas-deferens
- (4) Requirement of low temperature for spermatogenesis

138. Secretions from which one of the following glands are rich in fructose, calcium and some enzymes?
 (1) Salivary glands
 (2) Female sex accessory glands
 (3) Male sex accessory glands
 (4) Liver
139. Identify the correct sequence of events taking place in angiosperms.
 (a) Megasporogenesis
 (b) Development of embryo
 (c) Pollen-pistil interaction
 (d) Development of endosperm
 (e) Pollination
 (f) Entry of pollen tube into the ovule
 (g) Megagametogenesis
 (1) $a \rightarrow e \rightarrow g \rightarrow c \rightarrow f \rightarrow b \rightarrow d$
 (2) $a \rightarrow g \rightarrow c \rightarrow e \rightarrow f \rightarrow d \rightarrow b$
 (3) $a \rightarrow g \rightarrow e \rightarrow c \rightarrow f \rightarrow d \rightarrow b$
 (4) $g \rightarrow a \rightarrow e \rightarrow c \rightarrow f \rightarrow b \rightarrow d$
140. If for some reason, the vasa efferentia in the human reproductive system get blocked, the gametes will not be transported from:-
 (1) Testes to epididymis
 (2) Epididymis to vas deferens
 (3) Ovary to uterus
 (4) Vagina to uterus
141. Following are the steps involved in the process of embryo development in dicot?
 (a) Heart-shaped embryo
 (b) Mature embryo
 (c) Globular embryo
 (d) Zygote
 Choose the correct sequence of step of embryo development from the option given below :
 (1) $d \rightarrow c \rightarrow a \rightarrow b$
 (2) $c \rightarrow b \rightarrow d \rightarrow a$
 (3) $b \rightarrow c \rightarrow a \rightarrow d$
 (4) $a \rightarrow d \rightarrow b \rightarrow c$
142. Select the correct sequence of transport of sperm cells in male reproductive system:-
 (1) Testis $\rightarrow$ Epididymis $\rightarrow$ Vasa efferentia $\rightarrow$ Rete testis $\rightarrow$ Inguinal canal $\rightarrow$ Urethra
 (2) Seminiferous tubules $\rightarrow$ Rete testis $\rightarrow$ Vasa efferentia $\rightarrow$ Epididymis $\rightarrow$ Vas deferens $\rightarrow$ Ejaculatory duct $\rightarrow$ Urethra $\rightarrow$ Urethral meatus
 (3) Seminiferous tubules $\rightarrow$ Vasa efferentia $\rightarrow$ Epididymis $\rightarrow$ Inguinal canal $\rightarrow$ Urethra
 (4) Testis $\rightarrow$ Epididymis $\rightarrow$ Vasa efferentia $\rightarrow$ Vas deferens $\rightarrow$ Ejaculatory duct $\rightarrow$ Urethra $\rightarrow$ Urethral meatus
143. Given below are two statements:
Statement-I: When pollen grains are released from pollen sac of anther and they land on stigma of pistil, called pollination.
Statement-II: Pollen tube enters into embryo sac where two male gametes fuse with female gamete.
 In the light of the above statements, choose the most appropriate answer from the options given below:
 (1) Both statement -I and II are correct.
 (2) Statement-I is correct but statement-II is incorrect.
 (3) Statement-I is incorrect but Statement -II is correct.
 (4) Both Statement-I and Statement -II are incorrect.
144. Given below are two statements :
Statement-I: A typical anther is tetrasporangiate.
Statement -II: A typical anther is bilobed.
 In the light of the above statements, choose the most appropriate answer from the options given below:
 (1) Both statement -I and II are correct.
 (2) Statement-I is correct but statement-II is incorrect.
 (3) Statement-I is incorrect but Statement -II is correct.
 (4) Both Statement-I and Statement -II are incorrect.
145. Identify the correct set of statements –
 (a) In gymnosperms, seeds are not enclosed by fruit wall.
 (b) Seed is the end product of sexual reproduction.
 (c) Flower is the structure in which reproductive whorls of angiosperms are present.
 (d) Development of endosperm in angiosperm is post- fertilization event.
 Choose the correct answer from the given options below:
 (1) only b, c & d are correct
 (2) only b and c are correct
 (3) All a, b, c and d are correct
 (4) only a and c are correct
146. Read the given statements and select the statements which are correct :-
 (a) The uterus is single pair and is also called womb.
 (b) The shape of uterus is like an inverted pear.
 (c) The uterus open into vagina through a narrow cervix.
 (d) Birth canal is formed by cervical canal and vagina.
 (e) Perimetrium of uterus wall is outermost and thick glandular layer.
 (1) a, b & c
 (2) b, c & d
 (3) d & e
 (4) a & e

147. Consider the following four statements (a-d) and select the options, which includes all the correct ones only :
- Maturation phase is the longest phase in oogenesis process of human.
 - Oogenesis is initiated at puberty.
 - Primary oocyte temporarily arrested in prophase-I-stage.
 - During oogenesis meiotic-first division is an equal division leading to formation of secondary oocyte and polar body.
- statement (a), (b), (c)
 - statement (a), (c)
 - statement (b), (d)
 - statement (a), (d)
148. Which of the following is world's first non-steroidal pill developed by scientists at Central Drug Research Institute (CDRI) in Lucknow, India?
- i-Pill
 - Mala-D
 - Saheli
 - Mala-N
149. The signals for parturition originate from the :-
- Fully developed foetus
 - Placenta
 - Uterus
 - Ovary
- (a) & (b)
 - (a) & (d)
 - (c) & (d)
 - (b) & (d)
150. Arrange the following events of female reproductive cycle in correct sequence –
- Secretion of FSH
 - Growth of corpus luteum
 - Growth of follicles
 - Ovulation
 - Sudden increase in level of LH.
 - Formation of corpus albicans.
- III → I → IV → II → VI → V
 - I → III → V → IV → II → VI
 - I → IV → III → V → II → VI
 - II → I → III → IV → VI → V
151. Given below are two statements:
- Statement-I:** In adults, each testis is oval in shape, with a length of about 2 to 3 cm and a width of about 4 to 5 cm.
- Statement-II:** Each testis has about 250 compartment called testicular lobules.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- Both Statement -I and II both are incorrect.
 - Statement -I is correct but statement-II is incorrect
 - Statement-I is incorrect and Statement-II is correct.
 - Both Statement-I and Statement-II are correct.
152. Tubectomy is a method of sterilization in which:
- Small part of the fallopian tube is removed or tied up.
 - Ovaries are removed surgically
 - Small part of vas deferens is removed or tied up.
 - Uterus is removed surgically.
153. Given below are two statements:
- Statement-I:** The milk produced during the initial few days of lactation is called colostrum.
- Statement-II** The mammary glands of the female undergo differentiation during pregnancy and starts producing milk towards the end of pregnancy the process called lactation.
- Both Statement -I and II both are incorrect.
 - Statement -I is correct but statement-II is incorrect
 - Statement-I is incorrect and Statement-II is correct.
 - Both Statement-I and Statement-II are correct.
- Direction (Q.154 to Q.155):** Read the **assertion** and **reason** carefully to mark the correct option out of the options given below:
- If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
 - If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
 - If the assertion is true but the reason is false.
 - If both the assertion and reason are false
154. **Assertion:** Size of breasts increases at puberty in human female
- Reason:** Prolactin secretion starts at puberty
- A
 - B
 - C
 - D
155. **Assertion:** Corpus Luteum degenerates if the ovum remains unfertilized
- Reason:** Corpus Luteum secretes Progesterone which maintains the pregnancy
- A
 - B
 - C
 - D

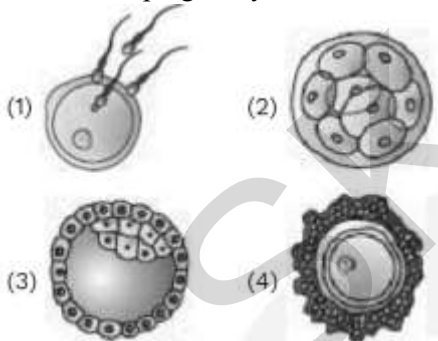
156. Identify the correct set of statements:

- (a) Each testicular lobule contains 1-3 highly coiled seminiferous tubules in which sperms are produced
- (b) The regions outside the seminiferous tubules called interstitial spaces, contains small blood vessels and interstitial cells or Leydig cells. or Leydig cells.
- (c) The male sex accessory ducts include testes, vasa efferentia, epididymis and vas deferens.
- (d) The enlarged end of penis called glans penis is covered by a tight fold of skin called fore skin.
- (e) The secretion of bulbourethral glands helps in the lubrication of the penis

Choose the correct answer from options given below:

- (1) (a) and (d) only
- (2) (a), (b) and (d) only
- (3) (a), (b), (c) and (e) only
- (4) (a), (b) and (e) only

157. Which of the following human developmental stage becomes embedded in the uterine endometrium by a process called implantation and leads to pregnancy?



158. During pregnancy, production of different hormones increases. This is essential for which of the following purposes

- (1) Supporting foetal growth
- (2) Supporting metabolic changes in mother
- (3) Maintenance of pregnancy
- (4) All of the above

159. Medical Termination of Pregnancy (MTP) is considered safe up to how many weeks of pregnancy?

- (1) Eight weeks
- (2) Twelve weeks
- (3) Eighteen weeks
- (4) Six weeks

160. Identify the correct set of statements:

- (a) Each ovary is about 2-4 cm is length
- (b) Each ovary is connected to the pelvic wall and uterus by tendons
- (c) Mons pubis is a cushion of fatty tissue covered by skin and public hair
- (d) The clitoris is a tiny finger like structure which lies at the upper junction of the two labia minora below the urethral opening
- (e) The glandular tissue of each breast is divided into 15-20 mammary lobes containing clusters of cells called alveoli

Choose the correct answer from options given below:

- (1) (a) and (e) only
- (2) (a), (b), (d) and (e) only
- (3) (a), (c) and (e) only
- (4) (a), (c), (d) and (e) only

161. What would happen to the levels of gonadotropins on surgical removal of ovaries?

- (1) Blood gonadotropins level will remains same
- (2) Blood gonadotropin levels will rise markedly
- (3) Blood gonadotropin levels will decrease
- (4) Blood gonadotropin levels will slightly increase

162. Transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum, but can provide suitable environment for fertilization and further development, this method called

- (1) GIFT
- (2) ICSI
- (3) IVF
- (4) ZIFT

163. identify the correct set of statements:

- (a) Each spermatogonium is diploid and contains 46 chromosomes
- (b) The spermatids are transformed into sperm by the process called spermiation
- (c) The acrosome is filled with enzymes that help fertilization of the ovum
- (d) Secretion of epididymis, vas deferens, seminal vesicle and prostate are essential for maturation and motility of sperms

Choose the correct answer from options given below:

- (1) (a), (c) only
- (2) (b), (c), (d) only
- (3) (a), (b), (c), (d) only
- (4) (a), (c), (d) only

164. In Zygote Intra Fallopian Transfer, 8-celled embryo is transferred into:

- (1) Uterus
- (2) Vagina
- (3) Fallopian tube
- (4) Vas deferens

165. Identify the incorrect set of statements:

- (a) At birth only 6000–80000 primary follicles are left in each ovary
- (b) Tertiary follicle is characterised by a fluid filled cavity called antrum
- (c) Oogenesis is initiated during embryonic development stage
- (d) Each primary oocyte gets surrounded by a layer of zona pellucida and is called the primary follicle

Chose the correct answer from options given below:

- (1) (a),(b) (2) (b),(c)
- (3) (a),(d) (4) (c), (d)

166. Mark the option with all correct statements

- (A) Number of chromosomes in a zygote is 46.
 - (B) Zygote formation usually occurs at isthmic part of fallopian tube
 - (C) Zygote is surrounded by zona pellucida.
- (1) A and B Only (2) A and C Only
 - (3) B and C Only (4) A, B and C

167. Select the incorrect match between hormone, its target and sources: –

	Hormone	Target	Source
(1)	FSH	Ovary	Anterior pituitary
(2)	LH	Graafian follicle	Anterior pituitary
(3)	Relaxin	Smooth muscles in endometrium of uterus	Ovary
(4)	Oxytocin	Myometrium of uterus	Hypothalamus

168. Which of the following is present in ovarian cortex at the time of birth of the infant?

- (1) First polar body
- (2) Primary oocyte
- (3) Secondary oocyte
- (4) Second polar body

169. “A” cells start division and enter in “B” stage of Meiotic division and get temporarily “C” at this state, called “D”. Identify A,B,C and D

- (1) A : oogonia, B: Metaphase I, C: Arrested, D: primary oocyte
- (2) A: oogonia, B: Anaphase I, C: Released, D: Secondary oocyte
- (3) A : oogonia, B: prophase I, C: Arrested, D: primary oocyte
- (4) A : oogonia, B: Telophase I; C :Released, D: Secondary oocyte

170. Select incorrect statement:

- (1) LH triggers ovulation in ovary
- (2) LH and FSH decrease gradually during the follicular phase
- (3) LH triggers secretion of androgens from the Leydig cells
- (4) FSH stimulates the Sertoli cells which help in spermiogenesis

171. Given below are four methods (A – D) and their modes of action (a–d) in achieving contraception. Select their correct matching from the four options that follow:

Method	Mode of Action
A. The pill	(a) Prevents sperms reaching cervix
B. Condom	(b) Prevents implantation
C. Vasectomy	(c) Prevents ovulation
D. LNG-20	(d) Semen contains no sperms

- (1) A-(c), B-(d), C-(a), D-(b)
- (2) A-(b), B-(c), C-(a), D-(d)
- (3) A-(c), B-(a), C-(d), D-(b)
- (4) A-(d), B-(a), C-(b), D-(c)

172. Here are some statements about recent amendment in MTP act, find the incorrect one:

- (1) It's intention was to reduce illegal abortions and consequent maternal mortality and morbidity.
- (2) It was amended in 2017
- (3) If pregnancy has lasted more than 24 weeks but less than 36 weeks, two registered medical practitioners must be of opinion that ground required for MTP exists
- (4) MTP can be performed if there is risk for pregnant woman or the child if born will suffer from physical or mental abnormality as to be seriously handicapped.

173. Identify the incorrect set of statements:

- (a) One ovum is released during the middle of each menstrual cycle
- (b) Menstrual phase lasts for 3-5 days
- (c) Lack of menstruation is surely indicative of pregnancy
- (d) The corpus luteum secretes large amount of progesterone which is essential for formation of endometrium
- (e) During pregnancy all events of the menstrual cycle stop and there is no menstruation

Choose the correct answer from options given below:

- (1) (a), (b), (e)
- (2) (c), (d)
- (3) (a), (c), (d)
- (4) (a),(b),(d)

174. Identify the correct set of statements:

- (a) A structural and functional unit between foetus and maternal body is called placenta
- (b) During pregnancy the level of cortisol, prolactin and thyroxine are increased several folds in the maternal blood
- (c) The inner calls mass contains stem cells which have the potency to give rise to all the tissues and organs
- (d) The doctor inject oxytocin to induce delivery
- (e) Colostrum contains several antibiotics

Choose the correct answer from options given below:

- (1) (a), (b), (c), (d) only
- (2) (a), (b), (d), (e) only
- (3) (a), (c), (c), (d) only
- (4) (b), (c), (d), (e) only

175. Which of the following ART is not used in IVF program?

- (1) ZIFT
- (2) GIFT
- (3) IUT
- (4) All of these

176. Which of the following is seen in pregnancy test

- (1) FSH in urine?
- (2) hCG in urine
- (3) Progesterone in urine
- (4) Gonadotropin in urine

177. Match the column-I with column-II

Column-I		Column-II	
A	Menstrual phase	i	Corpus luteum
B	Follicular phase	ii	Breakdown of Endometrium
C	Ovulatory phase	iii	Fully mature Graafian follicle
D	Luteal phase	iv	Release of ovum

- (1) A-iv, B-ii, C-i, D-iii
- (2) A-ii, B-i, C-iv, D-iii
- (3) A-ii, B-iii, C-iv, D-i
- (4) A-iv, B-i, C-ii, D-iii

178. Read the following statements and choose option having correct statements -

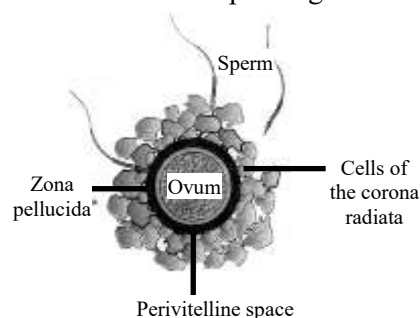
- (A) By the end of the second month of pregnancy the foetus develops limbs and digits.
- (B) By the end of 24th week the body is covered with fine hair.
- (C) The first sign of growing foetus maybe noticed by listening to the heart sound through the stethoscope.

- (1) (A), (B) Only
- (2) (A), (C) Only
- (3) (B), (C) Only
- (4) (A), (B), (C)

179. Which of the following statement is not correct?

- (1) Progestogens alone or in combination with estrogen can also be used by females.
- (2) Saheli is "once a week" pill.
- (3) Pills have to be taken daily for a period of 21 days.
- (4) Saheli has high side effect and low contraceptive value.

180. Study the given fig. and conclude the correct explanation from the options given below:



- (1) All the sperm attaches with ovum
- (2) Transport of ovum towards the sperm
- (3) Ovum and surrounded sperms were going to take part in fertilization
- (4) Sperm induces changes in the cells of corona radiata and blocks the entry of additional sperm

Answer Key and Solution

ANSWER KEY & SOLUTION

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	2	2	1	4	4	2	1	1	4	3	3	4	2	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	4	1	2	3	3	4	4	3	2	2	3	3	4	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	1	3	3	3	1	2	4	4	3	3	3	2	3	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	3	1	4	3	2	1	3	1	1	2	1	1	1	3
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	3	1	3	3	1	2	4	1	3	1	1	1	2	1	3
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	1	2	2	2	2	4	2	2	3	4	2	1	3	2
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	2	2	4	1	1	3	1	3	4	3	3	4	1	1	3
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	1	4	2	2	4	3	1	4	2	3	1	2	3	4	3
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	4	2	2	2	3	4	1	3	1	4	2	3	1	3	1
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	2	4	3	3	1	1	2	2	1	3	2	2	3	1	2
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	3	1	4	3	2	4	3	4	2	3	2	1	4	3	3
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	2	3	2	3	2	3	3	2	1	2	2	3	4	4	3

PHYSICS

Q.1 (4)

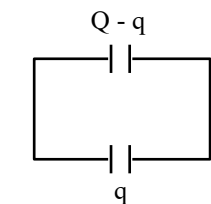
Both capacitors are in parallel

Q.2 (2)

$$KE = q(V_1 - V_2)$$

$$= 2 \times 1.6 \times 10^{-19} \times (70 - 50) = 40 \text{ eV}$$

Q.3 (2)

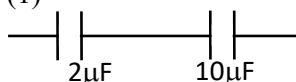


$$\frac{q}{C_2} = \frac{Q-q}{C_1}$$

$$\frac{q}{3C} = \frac{Q-q}{C} \Rightarrow q = 3Q - 3q$$

$$q = \frac{3Q}{4}, Q - q = Q/4$$

Q.4 (1)



$$V_{10\mu} = 12 \times \frac{2}{10+2} = 2V$$

$$U_{4\mu} = \frac{1}{2} \times 4\mu \times (2)^2 = 8\mu J$$

Q.5 (4)

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{Qr}{R^3} \Rightarrow E \propto \frac{1}{R^3}$$

Q.6 (4)

- (1)-The charge appearing on inner surface of B is $-Q$.
- (2)-The field between A and B is not zero.
- (3)-The charge appearing on outer surface of B is zero.

Initial condition:

Charger on outer surface of A = Q

Charger on inner surface of B = $-Q$

Charger on outer surface of B = Q

After earthing:

Charger on outer surface of A = Q

Charger on inner surface of B = $-Q$

Charger on outer surface of B = 0

$$\left[\text{since } V_B = \frac{K(Q)}{R} + \frac{K(-Q)}{R} = 0 \right]$$

Q.7 (2)

by conservation of mass: $n \frac{4\pi}{3} r^3 \rho = \frac{4\pi}{3} R^3 \rho$

Hence, $R = n^{\frac{1}{3}} r$.

Now, the capacitance of a sphere is proportional to its radius.

Hence the capacitance of the big drop will be C'

$$= n^{\frac{1}{3}} C$$

$$U_{\text{big}} = \frac{1}{2} n^{\frac{1}{3}} C V^2$$

$$U_{\text{small}} = \frac{1}{2} C V^2$$

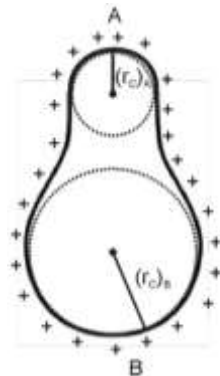
$$U_{\text{big}} : U_{\text{small}} = n^{1/3} : 1$$

Q.8 (1)

$$E_x = -\frac{\partial V}{\partial x}; E_y = -\frac{\partial V}{\partial y}; E_z = -\frac{\partial V}{\partial z}$$

Q.9 (1)

$$\Rightarrow E_A > E_B$$



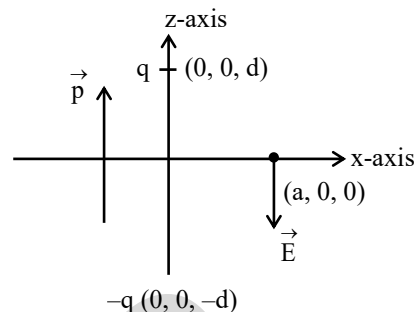
$$\sigma_A > \sigma_B \text{ as } (r_C)_A < (r_C)_B$$

$$E \propto \sigma \text{ (surface charge density)}$$

Q.10 (4)

Potential difference = 2(potential at the centre of one ring).

Q.11 (3)



Use formula $E = \frac{kp}{(r^2 + \ell^2)^{3/2}}$

$$\therefore E = \frac{1}{4\pi\epsilon_0} \frac{q \times 2d}{(a^2 + d^2)^{3/2}}$$

$$\therefore \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{2qd}{(a^2 + d^2)^{3/2}} (-\hat{k})$$

Q.12 (3)

$$\phi = \frac{q_{\text{in}}}{\epsilon_0} \text{ flux depend on the net charge enclosed}$$

by the Gaussian surface.

$$\text{So, } \boxed{\phi_1 = \phi_2}$$

Q.13 (4)

Imagine Gaussian cube of length L keeping outside q at the centre.

Q.14 (2)

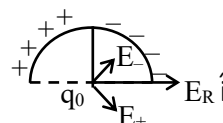
Electric field is non uniform in direction.

Q.15 (3)

$$E = \frac{KQ}{x^2} \Rightarrow E \propto \frac{1}{x^2}$$

At large distances the charged ring behaves as point charge.

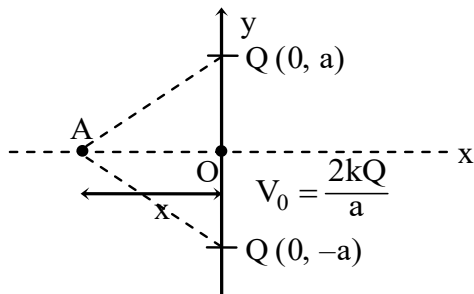
Q.16 (1)



Q.17 (4)

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

Q.18 (1)



Potential at centre is

$$V_0 = \frac{2kQ}{a}$$

Potential at axis

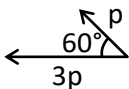
$$V_A = \frac{2kQ}{\sqrt{a^2 + x^2}}$$

At $x = \infty$, $V_\infty = 0$

$\therefore$ Graph will be 0 at $x = \pm \infty$ and will be maximum at $x = 0$

Q.19 (2)

Only two dipole are formed.



Q.20 (3)

$$\phi = E \cdot ds$$

$$\frac{N}{C} \cdot m^2 = \frac{N \cdot m \cdot m}{C} = \frac{J}{C} \cdot m = vm$$

Q.21 (3)

$$F_p = q \cdot E = mg \text{ and } F_q = qE > mg$$

Q.22 (4)

Electric field produces due to point charge is non-uniform.

Q.23 (4)

Total number of capacitors required will be

$$\frac{C_{eq}}{C} \times \left(\frac{V_{eq}}{V} \right)^2$$

Q.24 (3)

Q.25 (2)

Electric field of Q_0 terminated within the cavity and electric field due to Q charge is zero inside the cavity. Hence force on Q_0 is 0 but force on Q is not zero it is because of induce charge.

Q.26 (2)

Find equivalent capacitance of each and compare.

Q.27 (3)

Potential drop across C_1 is maximum

Q.28 (3)

$$V = \frac{W}{q_0}$$

Q.29 (4)

$$E_x = -\frac{\partial V}{\partial x}, E_y = -\frac{\partial V}{\partial y}, E_z = -\frac{\partial V}{\partial z}$$

Q.30 (1)

E is vector quantity so its direction will change.

Q.31 (1)

Electric field at the centre of the ring is zero.

Q.32 (1)

$$E_P = \frac{Q}{2\epsilon_0 A} + \frac{2Q}{2\epsilon_0 A} - \frac{3Q}{2\epsilon_0 A} = 0$$

Q.33 (3)

$$\epsilon_r = 81 \text{ for water}$$

Q.34 (3)

Resolve the forces; $F_R = 2 F \cos\left(\frac{\theta}{2}\right)$

$$\Rightarrow F_R = 2 \left(\frac{kq^2}{a^2} \right) \cos\left(\frac{120^\circ}{2}\right) = \frac{kq^2}{a^2}$$

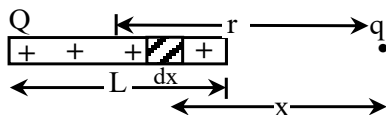
Q.35 (3)

$$a = \frac{qE}{m}$$

Q.36 (1)

Theory based

Q.37 (2)



$$dF = \frac{Kq\left(\frac{Q}{L}\right)dx}{x^2}$$

$$F = \frac{KQq}{L} \int_{r-\frac{L}{2}}^{r+\frac{L}{2}} \frac{dx}{x^2} = -\frac{KQq}{L} \left[\frac{1}{r+(L/2)} - \frac{1}{r-(L/2)} \right]$$

$$= \frac{KQq}{r^2 - (L/2)^2}$$

Q.38 (4)

$$E \propto \frac{1}{r^3}$$

Q.39 (4)

E.F. inside the charged solid conductor is zero.

Q.40 (3)

$$x = \frac{\sqrt{q_1}d}{\sqrt{q_1} - \sqrt{q_2}}; \quad d = 4 \text{ cm}$$

Q.41 (3)

$$\text{for } x < R : -E = \frac{9x}{3\epsilon_0} \Rightarrow E \propto x$$

$$\text{for } x > R : -E = \frac{kQ}{x^2} \Rightarrow E \propto \frac{1}{x^2}$$

Q.42 (3)

$$E = \frac{2k\lambda}{r}$$

Q.43 (2)

$$\phi = (\alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}) \cdot (\ell^2 \hat{j})$$

$$\phi = \ell^2 \beta$$

Q.44 (3)

Consider a cube of side 'a' keeping q at its centre.

Q.45 (1)

$$E = \frac{\sigma}{2\epsilon_0}$$

CHEMISTRY

Q.46 (1)

Sol. The reaction when KI is added in HgI_2



Since, number of ions decreased, so freezing point increases.

Q.47 (3)

Sol: $\Delta T_f = k_f \cdot m \cdot i$;

$$0.87 = 1.86 \times i \times \left(\frac{1}{345.6} \times \frac{1000}{25} \right); i = 4$$

Q.48 (1)

Sol. Molality of solution should be same

$$M_1 = M_2$$

$$\frac{4}{60} \times \frac{1000}{96} = \frac{10}{M_B} \times \frac{1000}{90} \text{ or } M_B = 160$$

Q.49 (4)

Sol: 40 mL O_3 dissolve in 100g water at 300 K and 1 atm
40 $\times$ 4 mL O_3 dissolve in 400 g water at 300 K and 1 atm

$\therefore M \propto P$ so $(40 \times 4) \times 4$ mL O_3 dissolve in 400g water at 300K and 4 atm

$$n_{\text{O}_3 \text{ dissolved}} = \frac{4 \times 640 \times 10^{-3}}{0.0821 \times 300} = 0.1$$

or mass of $\text{O}_3 = 4.8 \text{ g}$

Q.50 (3)

Sol: 1 mole Mn_3O_4 lose $\left(6 - \frac{8}{3} \right) \times 3 = 10$ mole e^-

So total charge required = $2 \times 10 = 20 \text{ F}$

Q.51 (2)

Sol: Let t sec. be used; No. of faradays = $\frac{4 \times t}{96500}$
 No. of moles of $H_2O = 4$, no. of equivalents of $H_2O = 4 \times 2$
 $\therefore \frac{4 \times t}{96500} = 4 \times 2$
 $\Rightarrow t = 1.93 \times 10^5$

Q.52 (1)

Sol: $E = E^\circ - \frac{0.059}{2} \log k$
 At equ. $E = 0$
 $E^\circ_{\text{cell}} = \frac{0.059}{2} \log k$
 $E^\circ_{\text{cell}} = \frac{0.059}{2} \times 12 = 0.354 \text{ volt}$

Q.53 (3)

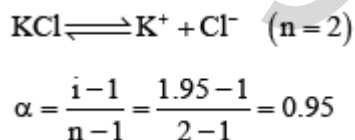
Sol: Reactivity of $Cu <$ Reactivity of Fe

Q.54 (1)

Sol: The solution shows negative deviation from Raoult's law because the vapour pressure of the solution is lower than that of either of two liquids.

Q.55 (1)

Sol:



Q.56 (2)

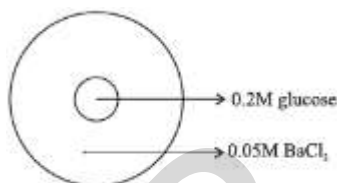
Sol: Low concentration of oxygen in the blood and tissues of people living at high altitude is atmospheric pressure. Because at high altitude, the partial pressure of oxygen is less than at the ground level. This decreased atmospheric pressure causes release of oxygen from blood.

Q.57 (1)

Sol: According to definition of osmotic pressure that $\pi = CRT$. For concentrated solution C has higher value than dilute solution.

Q.58 (1)

Sol:



$$\begin{aligned} \pi &= (C_2 - C_1)RT \\ &= (0.2 - 0.05 \times 3) \times 0.0821 \times 300 \\ &= (0.2 - 0.15) \times 0.0821 \times 300 \\ &= 0.05 \times 0.0821 \times 300 \\ &= 1.23 \text{ atm} \end{aligned}$$

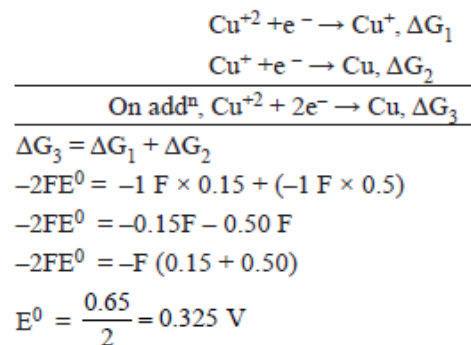
Q.59 (1)

Sol:

$$\begin{aligned} P^\circ - P_s &= 10 \\ \text{Apply, } \frac{P^\circ - P_s}{P^\circ} &= x_B \\ \frac{10}{P^\circ} &= 0.2 \Rightarrow P^\circ = 50 \\ \text{Then, } P^\circ - P_s &= 20 \\ \frac{P^\circ - P_s}{P^\circ} = x_B &\Rightarrow \frac{20}{50} = x_B = 0.4 \quad (\because x_A + x_B = 1) \\ \therefore x_A &= 0.6 \end{aligned}$$

Q.60 (3)

Sol:



Q.61 (3)

Q.62 (1)

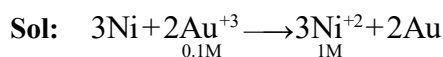
Sol:

$$\Lambda_{eq}^{\infty} = 200 \text{ ohm}^{-1} \text{cm}^2 \text{eq}^{-1} \quad \Lambda_{eq}^v = 10 \text{ ohm}^{-1} \text{cm}^2 \text{eq}^{-1}$$

$$\text{Degree of Ionisation} = \frac{\Lambda_{eq}^v}{\Lambda_{eq}^{\infty}} = \frac{10}{200} = 0.05$$

$$\text{Degree of ionisation (in\%)} = 0.05 \times 100 = 5\%$$

Q.63 (3)



$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{6} \log \frac{1^3}{0.1^2} = 1.77\text{V}$$

Q.64 (3)

Sol: Only Au^{3+} , Ag^{+} and Cu^{2+} will deposit at cathode.
 Li will not deposit at cathode because SRP of water is -0.8274V so after Cu^{2+} ; H_2 will evolve at cathode.

Q.65 (1)

Sol: No. of moles of glucose = $\frac{10}{180} = 0.0555$

$$\text{No. of moles of water} = \frac{90}{18} = 5$$

$$\text{No. of moles of solution} = 5.0555$$

$$\text{Mole fraction of glucose} = \frac{\text{No. of moles of Glucose}}{\text{No. of moles of solution}}$$

$$= \frac{0.0555}{5.0555} = 0.01$$

Q.66 (2)

Sol: If $p^{\circ} = 100\text{mm}$, $p_5 = 100 - 10 = 90\text{mm}$

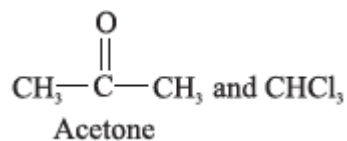
$$\frac{p^{\circ} - p_5}{p^{\circ}} = \frac{w_2 / M_2}{w_1 / M_1} = \frac{w_2}{M_2} \times \frac{M_1}{w_1}$$

$$\frac{100 - 90}{100} = \frac{w_2}{60} \times \frac{18}{180}$$

$$\text{or } w_2 = \frac{10}{100} \times 60 \times 10 = 60 \text{ g}$$

Q.67 (4)

Sol:



Acetone and chloroform will show a negative deviation due to their association after mixing.

Q.68 (1)

Q.69 (3)

Q.70 (1)

Sol.

Q.71 (1)

Sol.

Q.72 (1)

Sol.

Q.73 (2)

Sol.

Q.74 (1)

Sol.

Q.75 (3)

Sol.

Q.76 (2)

Sol.

Q.77 (1)

Sol.

Q.78 (2)

Sol.

Q.79 (2)

Sol. For an ideal solution $\Delta H_{\text{mix}} = 0$ and ΔS_{mix} is always positive so ΔG_{mix} is negative.

Q.80 (2)

Sol. FACT

Q.81 (2)

Sol. Ionic molar conductivity of H^+ is very high and NH_4OH is a weak electrolyte.

Q.82 (4)

$$\text{Sol. } n_{O_2} = \frac{V_{ml}}{22400} = \frac{5600}{22400} = \frac{1}{4}$$

$$w_{O_2} = n_{O_2} \times (M_w)_{O_2}$$

$$= \frac{1}{4} \times 32 = 8g$$

Faraday IInd law

$$\frac{w_{Ag}}{w_{O_2}} = \frac{E_{Ag}}{E_{O_2}} = \frac{108}{8}$$

$$\frac{w_{Ag}}{8} = \frac{108}{8}$$

$$w_{Ag} = 108$$

Q.83 (2)

$$\text{Sol. } G = \frac{1}{R} = \frac{1}{210} \text{ ohm}^{-1}$$

$$\kappa = \frac{1}{R} \cdot \frac{\ell}{A} = \frac{1}{210} \cdot \frac{0.84}{1} = 4 \times 10^{-3} \text{ ohm}^{-1} \text{ cm}^{-1}$$

$$\lambda_{eq} = \frac{\kappa \times 1000}{N} = \frac{4 \times 10^{-3} \times 1000}{0.01}$$

$$= 400 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$$

$$\lambda_m = \lambda_{eq} \times V_f = 400 \times 2 = 800 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

Q.84 (2)

Sol.

Q.85 (3)

Sol.

Q.86 (4)

Sol.

Q.87 (2)

Sol.

Q.88 (1)

Sol.

Q.89 (3)

Sol.

Q.90 (2)

Sol.